Ozgur Ural, Ph.D.

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Senior AI Engineer & Researcher | Ph.D. in ML Security & Distributed Systems

Senior software engineer and researcher with 11 + years of software engineering experience across cybersecurity, defense, and simulation systems. Holding a Ph.D. (August 2025) from Embry-Riddle Aeronautical University in ML Security and Distributed Systems, with peer-reviewed publications in IEEE Access on Proof-of-Learning and model watermarking.

My research and engineering work focuses on **secure, verifiable machine learning for critical infrastructure**: systems that must be performant, auditable, and secure. As U.S. federal agencies and industry accelerate AI adoption across energy, defense, and public administration, the need for engineers who can build and formally validate such systems is acute. With a record of IEEE publication, active journal refereeing, program committee membership, and over a decade of production engineering experience, I bring a rare combination of research depth and systems-level execution to this challenge.

TECHNICAL SKILLS

- **ML & AI:** PyTorch, TensorFlow, scikit-learn, federated learning, distributed ML training, model watermarking, Proof-of-Learning, adversarial ML, large-scale model evaluation
- **Languages:** C, C++, Java, Scala, Python, TypeScript/JavaScript
- **Distributed & Data:** gRPC/Protobuf, Redis, PostgreSQL, REST, real-time data pipelines
- **Systems:** Linux, multithreading, networking, real-time/low-latency, backpressure
- **Web & UI:** Svelte, HTML, CSS/Tailwind
- **DevOps:** Docker, GitLab CI/CD, Jenkins, Git

EXPERIENCE

Avion Full Flight Simulators — Amsterdam, Netherlands

Senior Software Engineer — Oct 2023 – Present

- Developed real-time simulator software using C, C++, and Scala.
- Built a low-latency real-time simulation data pipeline to receive telemetry, normalize/aggregate and serialize to JSON/Protobuf, cache in Redis, persist in PostgreSQL, and serve the frontend via gRPC/HTTP; sustained ~50 GB/s real-time ingest with batching and bounded-queue backpressure to keep dashboards responsive.
- Built web applications for the Avion Cloud Environment to monitor, configure, and diagnose core simulator components using TypeScript/JavaScript, Svelte, Scala, Python, gRPC & Protocol Buffers, HTML, and CSS/Tailwind.
- Containerized and deployed services with Docker; implemented caching and storage with Redis and PostgreSQL; automated testing and deployments via GitLab CI/CD.
- Translated requirements and specifications into efficient solutions, rapidly prototyping complex systems.
- Incorporated multithreading and efficient I/O (sync/async where appropriate) with bounded-queue backpressure to meet rigorous real-time requirements.

Embry-Riddle Aeronautical University, Daytona Beach — Florida, USA

Graduate Research Assistant in Electrical Engineering and Computer Science — Aug 2021 – Aug 2025

- Thesis Advisor: Dr. Kenji Yoshigoe
- Dissertation: "Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking"
- Conducted original research advancing the security of machine learning verification protocols, with direct applications to AI integrity in defense, finance, and critical infrastructure systems.
- Published three peer-reviewed articles in IEEE Access (impact factor: 3.9) as first author; work cited internationally and reviewed by leading experts in ML security and distributed systems.

Havelsan — Ankara, Turkey

Software Team Lead — Nov 2020 – Aug 2021

Havelsan DLP Data Leakage Prevention product.

- Technologies and tools used: C, C++, Java, Go, JavaScript, Qt, Sonar, Pardus, Windows, Git, Jira.
- Led a team of 14 engineers (developers, QA, DevOps, support).

- Developed and maintained key features of the endpoint agent, ensuring robust data protection.
- Coordinated with stakeholders to align project goals with business objectives, ensuring timely delivery.
- Provided mentorship and training to team members, fostering a collaborative work environment.

STM Defence Technologies Ankara, Turkey

Expert Software Engineer Feb 2019 – Nov 2020

Kargu Autonomous Rotary-Wing Attack UAV and Togan Autonomous Multi-Rotor Reconnaissance UAV.

- Technologies and tools used: C++11, Boost, Qt, QML, Google Test, Clang-Tidy, Ubuntu, Git.
- Contributed to the architecture and implementation of the mission-control unit, enhancing drone operational capabilities.
- Developed components of the ground-control unit using Qt and QML, ensuring effective communication and control.
- Designed and implemented Power-On and Continuous Built-In Tests to ensure system reliability and performance.
- Worked closely with cross-functional teams to align technical solutions with project requirements and military standards.

Comodo Cybersecurity Ankara, Turkey

Expert Software Engineer Jul 2014 – Feb 2019

Comodo Dome Secure Web Gateway, Comodo Patch Manager, and the Chromium-based web browser Comodo Dragon.

- Technologies and tools used: C++17, Boost, Qt, Git, Jira, Jenkins, WinAPI, Bitbucket, JavaScript, jQuery.
- Led the design and development of Windows services and applications to enhance functionality and reliability.
- Established unit testing and code-review protocols to ensure software quality.
- Pioneered the development of advanced browser features, enhancing user security and experience.
- Developed statistical reporting modules, providing product managers with insights for data-driven decision-making.
- Coordinated with cross-functional teams, including product management and QA.
- Implemented and maintained CI/CD pipelines using Jenkins.

EDUCATION

Embry-Riddle Aeronautical University, Daytona Beach Florida, USA

Ph.D. in Electrical Engineering and Computer Science 2021 – 2025 (conferred Aug 2025)

- Dissertation: "Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking"
- Advisor: Dr. Kenji Yoshigoe | Diploma verification available

Middle East Technical University (METU) Ankara, Turkey

Master of Science in Cyber Security 2015 – 2019

- Thesis: Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Turkish Newspaper Data

Middle East Technical University (METU) Ankara, Turkey

Bachelor of Science in Computer Engineering 2010 – 2014

- Institutional note: METU is widely regarded as Turkey's leading engineering university and is ranked 269th globally (QS World University Rankings 2026).

Anadolu University Eskisehir, Turkey (Distance/Online)

Bachelor of Business Administration 2012 – 2017

PUBLICATIONS

- Ural, O. and Yoshigoe, K. (2025). SecurePoL: Integration of Watermarking With Proof-of-Learning to Enhance Security Against Spoofing Attacks. *IEEE Access*, vol. 13, pp. 213067–213091. DOI: 10.1109/ACCESS.2025.3642198.
- Ural, O. (2025). Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Doctoral dissertation, Embry-Riddle Aeronautical University, Daytona Beach, Florida.

- Ural, O. and Yoshigoe, K. (2024). Enhancing Security of Proof-of-Learning against Spoofing Attacks using Feature-Based Model Watermarking. *IEEE Access*. DOI: 10.1109/ACCESS.2024.3489776.
- Ural, O. and Yoshigoe, K. (2023). Survey on Blockchain-Enhanced Machine Learning. *IEEE Access*, pp. 145331–145362. DOI: 10.1109/ACCESS.2023.3344669.
- Ural, O. and Acartuurk, C. (2021). Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. Proceedings of the 7th International Conference on Information Systems Security and Privacy (ICISSP), ISBN 978-989-758-491-6; ISSN 2184-4356, pp. 66–76. DOI: 10.5220/0010201600660076.
- Ural, Ozgur and Erdur, Efe. (2016). Secure Proxy on Cloud. DOI: 10.13140/RG.2.2.24058.08649.
- Ural, Ozgur (2014). Autonomous Cargo and Mail Delivery. Turkish Autonomous Robots Conference, Ankara, Turkey.

PROFESSIONAL SERVICE

For an up-to-date list of publications and citations, please see my Google Scholar profile.

- **Journal Referee:** IEEE Access and other peer-reviewed journals in security, privacy, machine learning, distributed systems, and blockchain. Regularly invited by editors to assess the validity, novelty, and technical rigor of original research submissions.
- **Reviewer:** IEEE, ACM, and other international conferences/symposia in security, privacy, machine learning, distributed systems, and blockchain.
- **Program Committee Member (Summer 2026):** 2nd Workshop on NLP Applied to Information and Cyber Security (NLPAICS 2026), University of Alicante, Spain. Selected by workshop organizers based on recognized expertise.
- **Session Chair:** Sessions aligned with research expertise in ML security and distributed systems.

AWARDS & HONORS

- The Scientific and Technological Research Council of Turkey (TUBITAK) National Project Competition – First Prize, June 2014.
- Middle East Technical University Graduation Projects Competition – Second Prize, June 2014.